FINANCE 1 FOR BUSINESS · UVA

Week-by-week summary

Weeks 1–3, everything for the midterm: summaries per chapter, every formula marked as given on the formula sheet or learn by heart, the exam traps that cost the most points, and key terms.

Week 1	Corporation, statements, arbitrage · Ch 1, 2, 3 + Hart & Zingales (2017)
Week 2	Time value, interest rates, bonds · Ch 4, 5, 6
Week 3	Investment rules, capital budgeting · Ch 7, 8 + Graham & Harvey (2022)

How to use this

1. Read one week's summary (about 20 minutes), then do that week's practice questions on the site.
2. For every mistake, find the matching line under *Exam traps*.
3. From day 3 on, do the past papers and mock midterms under exam conditions with only the formula sheet.

WEEK 1

Corporation, statements, arbitrage

Ch 1, 2, 3 + Hart & Zingales (2017)

Ch 1 · The corporation

A **sole proprietorship** (eenmanszaak) has one owner who is also manager and is personally liable. In a **partnership** (VOF, maatschap, CV) all partners are personally liable. An **LLC** (BV) and a **corporation** (NV) are legal entities: shareholders are **not** liable for the firm's debts, and ownership is separated from control. NV shares trade on an exchange, BV shares do not. Corporate profit is taxed twice: corporate tax, then income tax on dividends.

Objective: maximize **shareholder value** (the value of the shares), not profit. Shareholders are the residual claimants (last in line), so in theory they care about everyone else too. This assumes no agency problems and that externalities are priced or regulated.

Agency problem: managers (agents) can pursue their own utility instead of shareholders' (principals'), e.g. Tyco's CEO Kozlowski. Remedies: board oversight, pay tied to performance, product-market competition, and takeovers after the share price falls.

Trading: you buy at the **ask**, sell at the **bid**; the **bid-ask spread** is the market maker's profit.

Ch 2 · Statements, market vs book value

Balance sheet: Assets = Liabilities + Equity. **Income statement:** Sales − COGS = gross profit; − general costs − depreciation = operating income = **EBIT**; − interest = pretax income (EBT); − taxes = **net income**.

Book value is backward-looking (historical cost, accounting rules). **Market value** is forward-looking (expected cash flows, risk, sentiment). Market cap = share price × shares. Market-to-book = market value of equity ÷ book value of equity. Example: Firm Z, factory cost $100, pays $150 in 1 year at 8%: assets are worth 150/1.08 = $139, so equity = 139 − 40 = $99, share price $9.90, M/B = 99/60 = 1.65. A project with NPV $5 raises equity to $104 and the price to $10.40.

Enterprise value = market cap + debt − cash = value of the operating assets. Net debt = debt − cash.

Ratio group	On midterm formula sheet	Learn by heart (not given)
Liquidity, coverage, leverage	Current ratio, EBIT/interest, debt-to-capital	Debt-equity (book, market), debt-to-EV, net profit margin
Working capital	AR days, AP days, inventory days	–
Returns, valuation	ROE	M/B, P/E, EV/Sales, EV/EBITDA

The crossed-out ratios on the course list (margins except net, quick, cash, equity multipliers, EV/EBIT, asset turnover, ROA, ROIC, EBITDA coverage) are excluded.

Profit versus cash flow

Profit follows accounting rules; cash flow follows money. Revenues are not always inflows and costs not always outflows. **Cash flow = profit − increase in NWC**, where NWC = current assets − current liabilities (here accounts receivable − accounts payable). Example: sales 100 (80 received), COGS 70 (60 paid): profit 30, cash flow 20, increase in NWC 10.

- Taxes are based on profit: 30% of EBIT 30 = 9, so cash flow = 21 − 10 = 11.

- **Capex** is a cash outflow but not a cost. **Depreciation** is a cost but not a cash outflow. It creates a **depreciation tax shield** (tax rate × depreciation).
- Inventory purchases are cash outflows that are not in profit.

FCF = unlevered net income + depreciation – capex – increase in NWC. Example: 15.4 + 8 − 24 − 10 = −10.6. Free cash flow excludes financing flows (separation principle); unlevered net income ignores interest.

Ch 3 · Valuation, arbitrage, law of one price

Use **market prices** to get cash values, but only in **competitive markets** (same price to buy and sell, ignoring transaction costs). A euro today beats a euro next year because you can invest it. Compare only at the **same moment**: **PV** = CF/(1+r) (discounting), **FV** = CF×(1+r) (compounding). Value a stream by adding the PVs.

NPV = PV(benefits) − PV(costs). Take the highest-NPV alternative; accept NPV > 0. Invest 8 to get 9 at 10%: NPV = −8 + 9/1.10 = 0.18 > 0, so accept.

Arbitrage: profit without risk or net investment (rare). **No-arbitrage price** = PV of all cash flows. A **normal market** has no arbitrage. Then the **law of one price** holds (equivalent assets trade at one price), which gives **value additivity**: Price(A+B) = Price(A) + Price(B), and firm value V = D + E. It also gives the **separation principle**: buying or selling securities has NPV 0, so investment decisions can be separated from financing decisions.

Bond arbitrage method: match the coupon bond's cash flows with zero-coupon bonds and compare prices. Zeros at 972.14, 939.62 and 906.24: the 10% coupon bond is worth 0.1×972.14 + 0.1×939.62 + 1.1×906.24 ≈ $1188, but trades at $1183.95. Buy the cheap bond, sell the replicating zeros.

Article · Hart & Zingales (2017)

Friedman argued managers should make as much money as possible within the rules. Hart and Zingales say **shareholder welfare ≠ market value**: shareholders are prosocial and care about more than money. Friedman's **separability** assumption (let companies make money and let people give to charity) fails when profit and damage are inseparable, e.g. oil digging, or Walmart selling high-capacity magazines. Politics also does not fix everything.

Amoral drift: hostile bidders only care about money, and dispersed shareholders tender anyway (collective action problem), so takeovers push firms from clean to dirty. There is no mirror-image drift, because nobody gains by buying a dirty firm to clean it. **Voting** by shareholders avoids this asymmetry. Authors' proposal: managers and asset managers follow investors' preferences, via votes or polls. Fiduciary duty is not just maximizing value (the Shkreli example). Prefer **invest and engage** over divestment.

Formulas

What	Formula	Exam
Current ratio	`Current assets / Current liabilities`	given
EBIT / interest	`EBIT / Interest expense`	given
Debt-to-capital	`Debt / (Equity + Debt)`	given
ROE	`Net income / Book equity`	given
AR days	`AR / Avg daily sales`	given
AP days	`AP / Avg daily COGS`	given
Inventory days	`Inventory / Avg daily COGS`	given
Free cash flow	`UNI + Depr - CapEx - ΔNWC`	given
FV	`CF × (1+r)^n`	given
PV	`CF / (1+r)^n`	given
Enterprise value	`Mkt cap + Debt - Cash`	learn
Market cap	`Price × Shares`	learn
Market-to-book	`Mkt equity / Book equity`	learn
P/E	`Price / EPS`	learn
Net profit margin	`Net income / Sales`	learn
Debt-equity	`Debt / Equity (book or market)`	learn
NPV	`-Investment + Σ CF/(1+r)^t`	learn
Cash flow (simple)	`Profit - increase in NWC`	learn
No-arbitrage price	`PV of all the security's cash flows`	learn

Exam traps

- ! EV uses **interest-bearing debt**, not total liabilities: accounts payable are not debt.
- ! Market-to-book = market value of equity / **book** equity. Book-to-market is the inverse.
- ! AR days divide by daily **sales**; inventory days and AP days divide by daily **COGS**.
- ! A ratio can fall while the level rises: net income up, net profit margin down when sales grow faster.
- ! When you **sell** you get the **bid**; when you buy you pay the **ask**.
- ! Arbitrage: buy the cheap side, sell the expensive side, and the future cash flows must cancel exactly.
- ! Depreciation is a cost but not a cash outflow: add it back for cash flow.
- ! Shareholder **value** (Friedman) ≠ shareholder **welfare** (Hart & Zingales) ≠ profit.

Key terms

Sole proprietorship One owner who is also the manager; personal liability.

Partnership Several owners, all personally liable (VOF, maatschap).

Limited liability Shareholders lose at most their investment, not the firm's debts.

Shareholder value The value of the firm's shares; the standard objective of the corporation.

Residual claimant	Shareholders are paid last, after creditors.
Agency problem	Managers pursue their own interests instead of the shareholders'.
Externality	A cost or benefit to others that is not priced (e.g. pollution).
Bid-ask spread	Ask price minus bid price; the market maker's profit.
EBIT	Earnings before interest and taxes = operating income + other income.
Market capitalization	Share price × number of shares = market value of equity.
Market-to-book ratio	Market value of equity ÷ book value of equity.
Enterprise value	Market cap + debt − cash; market value of operating assets.
Net working capital	Current assets − current liabilities.
Free cash flow	Unlevered net income + depreciation − capex − increase in NWC.
Depreciation tax shield	Tax saved because depreciation is deductible: tax rate × depreciation.
Discount factor	1/(1+r): converts a future cash flow into present value.
NPV	PV of all cash flows including the initial investment.
Arbitrage	Risk-free profit with no net investment.
Law of one price	Equivalent assets in competitive markets trade at the same price.
Value additivity	Price of a bundle = sum of the prices of its components.
Separation principle	Securities trades have NPV 0, so investment and financing decisions are separate.
Amoral drift	Hart & Zingales: takeovers push companies toward ignoring externalities.

WEEK 2

Time value, interest rates, bonds

Ch 4, 5, 6

Ch 4 · Time value of money

$1 now is worth more than $1 later because of **impatience** (people prefer consumption now), **inflation** (money loses value) and **risk** (future cash flows are uncertain and people are risk averse).

Three rules of time travel: (1) only values at the **same point in time** can be compared or combined; (2) to move a cash flow forward, **compound** it: FV = C × (1+r)^n; (3) to move it backward, **discount** it: PV = C / (1+r)^n. Example at r = 5%: 1,000 today = 1,050 in a year. Both ways give the same answer, there is no preference. A stream is valued by adding the PV of each cash flow, and **NPV** > 0 means invest.

Ch 4 · Perpetuities, annuities, loans, IRR

Perpetuity (constant C forever): PV = C/r. **Growing perpetuity**: PV = C/(r − g), only valid if **g < r**. **Annuity** (constant C for N periods): PV = C/r × (1 − 1/(1+r)^N). **Growing annuity**: PV = C/(r − g) × (1 − ((1+g)/(1+r))^N). **FV of an annuity** = PV × (1+r)^N. Example: $2m growing 4% for 10 years at 11%: PV = 2/0.07 × (1 − (1.04/1.11)^10) ≈ $13.68m.

Early and late annuities: the formula assumes the first payment at time 1. If it starts at time 0 (early), multiply by (1+r), because every payment arrives one period sooner. If it starts at time 2 (late), divide by (1+r). The same adjustment works for perpetuities: with C = 10 and r = 2%, first payment at time 1 gives 500, at time 0 gives 510, first payment at time 2 gives 500/1.02 ≈ 490.20.

Loan payment (solve for C): C = P × r / (1 − (1+r)^−N). A $100,000 loan at 8% over 10 years costs $14,903 per year. **IRR** is the rate that makes NPV = 0: turning 1,000 into 2,000 in 6 years gives (2)^(1/6) − 1 = 12.25%.

Ch 5 · APR, EAR and matching the period

Key rule: change the interest rate to match the cash flow period, never the cash flows to match the rate. **APR** is simple interest per year without compounding: 1% per month is 12% APR with monthly compounding. **EAR** includes compounding: EAR = (1 + APR/k)^k − 1, so 1% per month = 1.01^12 − 1 = 12.68%. APR is a red flag: it cannot be used directly for PV or FV.

Converting: APR to period rate: divide by k (9% APR, quarterly → 2.25%). EAR to period rate: (1+EAR)^(1/k) − 1 (9% EAR → 1.09^(1/4) − 1 = 2.18% per quarter, 0.72% per month).

Four steps for a time-value question: (1) which cash flow form (annuity, perpetuity…); (2) which rate matches the cash flow period; (3) how many periods; (4) PV or FV. Example: $5 per month for 18 years at 6% APR: monthly rate 0.5%, N = 216, PV = 659.49, FV = 659.49 × 1.005^216 = 1,936.77. An **annuity (amortizing) loan** has equal payments, each part interest and part repayment, and the loan is fully repaid at time N. A **fair loan** has a balance equal to the PV of its payments: that PV is what you can borrow.

Ch 5 · What determines interest rates

Nominal rates are the quoted ones and discount nominal cash flows. **Real rate:** (1 + r) = (1 + rr)(1 + i), so rr ≈ r − i. Use nominal with nominal, or real with real. 6% nominal with 2% inflation gives 3.92% real.

Term structure / yield curve: the interest rate depends on maturity, and the yield curve plots EAR against term. It is usually upward sloping (normal); it is **inverted** when short rates are above long rates, which signals that investors expect lower future rates. Tutorial: one-year rates 2%, 1%, 0%, 0% give EARs of 2%, 1.499%,

0.997% and 0.75%. Match the term: discount a year-3 cash flow at the 3-year rate (10/1.00997^3 = $9.71, same as 10/(1.02 × 1.01 × 1.00)).

Taxes: use the after-tax rate, r × (1 − τ). **Risk:** riskier cash flows get a higher discount rate; use expected cash flows. The **opportunity cost of capital** is the best expected return available on an investment of comparable risk and term, and it is the discount rate you should use.

Ch 6 · Bonds, prices and yields

A **bond** is a tradeable debt contract. The issuer receives money now and promises coupons plus the **face value** at maturity; the holder carries the default risk. The coupon payment is CPN = coupon rate × face value / k. The **coupon rate only sets the cash flows**, it is never the discount rate. Equity, in contrast, promises no fixed cash flows. *Advantages of bonds:* easy external capital, cheaper than bank loans for large amounts, no voting rights, tax shield. *Disadvantages:* emission costs, fixed interest can trigger bankruptcy, default risk for the holder.

Prices: zero-coupon P = FV/(1+YTM)^n. Coupon bond P = CPN × annuity factor(y, N) + FV/(1+y)^N. **YTM** is the IRR of the promised cash flows, quoted as an APR (5% per 6 months = 10%). Price < face = **discount** bond, = face = **par**, > face = **premium**. Example: 6% coupon, semi-annual, 10 years, YTM 7.5% APR: 3.75% per period, 20 periods, P = 895.78. A zero priced at 96,618.36 for 100,000 in one year has YTM = 100,000/96,618.36 − 1 = 3.5%.

Rates and prices move opposite: when yields rise, prices fall, and the effect is bigger for longer maturities (**interest rate risk**). Tutorial: the 5-year, 6% semi-annual bond at 3% YTM costs 1,138.33; at 3.5% it costs 1,113.77; at 1,200 the YTM is 0.90% per half-year (1.80% APR). A bond bought at its fair price has NPV = 0, otherwise investors would push the price until it does.

Ch 6 · Bond arbitrage and credit risk

Replicate a coupon bond with zero-coupon bonds and compare prices (law of one price). Method 1, matching cash flows: a 10% annual, 3-year bond equals 0.1 of the 1-year zero, 0.1 of the 2-year zero and 1.1 of the 3-year zero: 0.1 × 972.14 + 0.1 × 939.62 + 1.1 × 906.24 = 1,188.04. If it trades below that, buy the bond and sell the zeros; if above, do the opposite. Method 2, yields: find each zero's YTM and discount the coupon bond's cash flows with them (book example: 100/1.035 + 100/1.04^2 + 1,100/1.045^3 ≈ 1,153).

Credit risk: a bond's promised yield (YTM, based on promised cash flows) is higher than its **expected return** (based on expected cash flows) when default is possible. The **yield spread** (credit or default spread) = corporate yield − Treasury yield: 3% − 2% = 1%. Ratings run from AAA to D; below BBB (Baa) is **speculative grade (junk)**. Quoted prices are **clean**; dirty (invoice) price = clean + accrued interest.

Formulas

What	Formula	Exam
PV perpetuity	`C / r`	given
PV growing perpetuity	`C / (r - g), g < r`	given
PV annuity	`C/r × (1 - 1/(1+r)^N)`	given
PV growing annuity	`C/(r-g) × (1 - ((1+g)/(1+r))^N)`	given
EAR from APR	`(1 + APR/k)^k - 1`	given
Zero-coupon bond price	`FV / (1+YTM)^n`	given
Coupon bond price	`CPN/y × (1 - 1/(1+y)^N) + FV/(1+y)^N`	given
FV annuity	`PV annuity × (1+r)^N`	learn
Early annuity (t = 0)	`PV annuity × (1+r)`	learn
Period rate from APR	`APR / k`	learn
Period rate from EAR	`(1+EAR)^(1/k) - 1`	learn
Loan payment	`C = P × r / (1 - (1+r)^-N)`	learn
IRR (one cash flow)	`(FV/PV)^(1/n) - 1`	learn
Real rate	`(1+r)/(1+i) - 1 ≈ r - i`	learn
After-tax rate	`r × (1 - tax rate)`	learn
Coupon payment	`CPN = coupon rate × FV / k`	learn
Multi-year EAR	`[(1+r1)(1+r2)…(1+rn)]^(1/n) - 1`	learn
Yield spread	`Corporate yield - Treasury yield`	learn

Exam traps

- ! An annuity or perpetuity formula gives the value **one period before** the first payment.
- ! First payment today? Multiply the ordinary annuity by (1 + r).
- ! Match the rate to the period: monthly cash flows need a monthly rate. From an EAR: (1+EAR)^(1/12) − 1; from an APR: APR/12.
- ! The EAR includes compounding, the APR does not. Never put an EAR into APR/k.
- ! Exact real rate: (1 + nominal)/(1 + inflation) − 1, not nominal − inflation.
- ! Coupon rate > YTM → premium; = → par; < → discount. Semi-annual bond: halve the coupon and the yield, double N.
- ! Long bonds (and zeros) are most sensitive to rate changes (interest rate risk).
- ! With default risk, YTM (promised) > expected return.
- ! Use each maturity's own rate from the yield curve, not the longest or an average rate.

Key terms

Time value of money — A positive interest rate makes money now worth more than the same amount later.

Compounding — Moving a cash flow forward in time: FV = C × (1+r)^n.

Discounting	Moving a cash flow backward in time: PV = C / (1+r)^n.
Perpetuity	Constant cash flow C forever; PV = C/r.
Growing perpetuity	Cash flow growing at g forever; PV = C/(r − g), needs g < r.
Annuity	Constant cash flow for a limited number of periods.
Early / late annuity	First payment before (time 0) or after (time 2) the normal time 1; multiply or divide the PV by (1+r).
IRR	Interest rate that makes the NPV of the cash flows zero.
Amortizing (annuity) loan	Equal periodic payments, each with interest and repayment; fully repaid at time N.
Fair loan	Loan balance equals the PV of the loan payments.
APR	Simple annual rate without compounding; 1% a month = 12% APR.
EAR	Annual rate including compounding: (1 + APR/k)^k − 1.
Nominal vs real rate	Nominal includes inflation; real is corrected for it: (1+r) = (1+rr)(1+i).
Term structure	The relationship between investment term and interest rate.
Yield curve	Graph of the term structure; inverted when short rates exceed long rates.
Opportunity cost of capital	Best expected return on an investment of comparable risk and term; the discount rate.
Face value (par)	Amount repaid to the bondholder at maturity.
Coupon rate	Sets the coupon payment (rate × face value); not a discount rate.
Zero-coupon bond	Bond with one payment, the face value at maturity; trades at a discount.
Yield to maturity (YTM)	IRR of a bond's promised cash flows; quoted as an APR.
Discount / par / premium bond	Price below / equal to / above face value.
Interest rate risk	Bond prices fall when rates rise, more so for long maturities.
Yield spread	Corporate bond yield minus Treasury yield (credit or default spread).
Junk bond	Speculative-grade bond rated below BBB (Baa).
Clean / dirty price	Clean is the quoted price; dirty = clean + accrued interest.

WEEK 3

Investment rules, capital budgeting

Ch 7, 8 + Graham & Harvey (2022)

Ch 7 · NPV and the IRR rule

NPV = $-I_0$ + PV(CF). Accept if NPV > 0: what you receive is worth more than what you invest, and firm value rises by the NPV today. Invest $100 for $120 in a year at 10%: NPV = −100 + 120/1.10 = $9.09. Perpetuity project (invest 250, 35 per year forever, r = 10%): NPV = −250 + 35/0.10 = 100.

IRR = the discount rate that makes NPV = 0 (like a bond's YTM). Accept if IRR > opportunity cost of capital. Invest 100, get 120: IRR = 20%. Perpetuity: IRR = C/I (35/250 = 14%). Growing perpetuity: IRR = C_1/I + g (25/400 + 1% = 7.25%). IRR − cost of capital is the maximum error in your cost-of-capital estimate that leaves the decision unchanged (14% − 10% = 4 points).

Ch 7 · When the IRR rule fails

IRR works for stand-alone projects where **all negative cash flows come before the positive ones**: NPV is then a smoothly **decreasing** function of the discount rate. It can disagree with NPV in three cases:

- **Delayed investment** (benefits first, costs later). Book deal: +1,000,000 now, −500,000 at t = 1, 2, 3. NPV at 10% = −243,426 (reject), but IRR = 23.38% > 10% (accept). NPV is **increasing** in r, so the rule is reversed.
- **Multiple IRRs** (cash flows change sign more than once). Book deal plus: +550,000, −500,000 ×3, +1,000,000 at t = 4 has IRRs of 7.164% and 33.673%. NPV is negative between them, −10,413 at 10%: reject.
- **No IRR**: NPV is always positive or always negative.

Most important lesson: **when in doubt, use NPV**.

Ch 7 · Payback, ROCE and Graham (2022)

Payback period = time to recover the initial investment. Accept if payback < the **cutoff period**. Invest 120, receive 30 per year: payback 4 years, so reject with a 3-year cutoff. Invest 250, receive 35 per year: 250/35 = 7.14 years.

Drawbacks: ignores time value of money, cost of capital and risk; ignores cash flows after the payback period; the cutoff is ad hoc. **Why used:** simple, fine for projects with similar risk and timing, and for liquidity-constrained (often small) firms. **ROCE / accounting rate of return** = average annual accounting profit ÷ investment, accept if above a hurdle rate; it uses accounting data, no risk or time-value correction.

Graham (2022) survey: NPV and IRR are used by roughly 75–80% of firms, but payback is still used by a large share (roughly 60–65%, more in small firms), despite its shortcomings.

Ch 7 · Choosing between projects, resource constraints

Mutually exclusive (at most one project): choose the highest **NPV**. Highest IRR can mislead when projects differ in **scale** (500% on €1 vs 20% on €1 million), **timing** (50% return for 1 year vs 5 years: NPV 36.36 vs 371.51 at 10%) or **risk**. Book store (invest 300,000, 63,000 growing 3%, r = 8%): IRR 24%, NPV = 1,260,000 − 300,000 = 960,000. Coffee shop (400,000, 80,000): IRR 23%, NPV = 1,600,000 − 400,000 = 1,200,000. Choose the coffee shop.

Resource constraints (not all positive-NPV projects fit): (1) check every feasible combination and take the highest total NPV, or (2) **profitability index** PI = NPV / resource consumed, invest in the highest PI first. PI is

fast but not always right. Budget 20: PI picks A+B+C (NPV 31.3) but A+B+D gives 34.

Ch 8 · Incremental earnings and unlevered net income

Incremental earnings = earnings with the project − earnings without it. Items that are the same either way are ignored.

- **Include:** extra revenues and costs, depreciation, **opportunity costs** (value of a resource in its best alternative use, e.g. lost rent), **cannibalization** (lost sales of existing products), salvage and terminal value.
- **Exclude: sunk costs** (paid regardless: past R&D, a completed feasibility study), fixed overhead that does not change, and **interest** (financing is judged separately: separation principle).

Unlevered net income = EBIT × (1 − τc) = (Revenues − Costs − Depreciation) × (1 − τc). HomeNet year 1: (26,000 − 11,000 − 2,800 − 1,500) × 0.6 = 6,420 ($000s). Cannibalization: 25% × 100,000 × $100 = $2.5m lost sales and 25% × 100,000 × $60 = $1.5m lost costs, net $1.0m pre-tax.

Ch 8 · Free cash flow, NWC and NPV

Profit ≠ cash flow. Corrections: credit sales and purchases (**ΔNWC**), depreciation (non-cash: add back, but it still saves tax = τc × depreciation, the **depreciation tax shield**), **CapEx** (cash out, not a cost). New loans and equity are financing and are ignored.

FCF = UNI + Depreciation − CapEx − ΔNWC. NWC = current assets − current liabilities = cash + inventory + receivables − payables. An increase in NWC is a cash outflow: NWC of 130 → 150 means −20. NWC is recovered at the end, so its cash flows sum to zero but still affect NPV through timing. Example: (2,400 − 900 − 480) × 0.75 + 480 − 600 − 120 = 525.

HomeNet FCF: −16,500; 5,100; 7,200; 7,200; 7,200; 2,700. At 12%: NPV = $5,027 ($000s), IRR = 24.1%.

Ch 8 · Analysing the project, two-stage valuation

Break-even analysis: the input level where NPV = 0 (for the discount rate it is the IRR). Invest 50, 10 forever, r = 5%: NPV = 150, IRR = 10/50 = 20%, break-even investment = 200, break-even cash flow = 50 × 0.05 = 2.5. **Sensitivity analysis:** change one assumption at a time (best/worst case) to see which matters most. **Scenario analysis:** change several at once (e.g. price and volume together).

Two-stage valuation: dividends of $2 for 5 years, then growing 2% forever, r = 7%. Stage 1 is an annuity: $2/0.07 \times (1 - 1/1.07^5) = 8.20$. Stage 2 is a growing perpetuity valued at t = 5: 2.04/(0.07 − 0.02) = 40.80, discounted: $40.80/1.07^5 = 29.09$. Price = 37.29.

Formulas

What	Formula	Exam
Free cash flow	`UNI + Depr - CapEx - ΔNWC`	given
PV perpetuity	`C / r`	given
PV growing perpetuity	`C / (r - g), g < r`	given
PV annuity	`C/r × (1 - 1/(1+r)^N)`	given
NPV	$-I_0 + \Sigma\ CF_t / (1+r)^t$	learn
IRR	`rate where NPV = 0`	learn
IRR (perpetuity)	C / I_0 (growing: $C_1/I_0 + g$)	learn
Payback period	years until cumulative CF ≥ I_0	learn
Profitability index	`NPV / resources consumed`	learn
Unlevered net income	`(Rev - Costs - Depr) × (1 - τc)`	learn
Depreciation tax shield	`τc × Depreciation`	learn
NWC	`Cash + Inventory + Receivables - Payables`	learn
Cash flow from NWC	`-ΔNWC`	learn
Break-even	`level of input where NPV = 0`	learn

Exam traps

- ! Delayed investment (+ first, − later): the IRR rule reverses. Compute NPV.
- ! Mutually exclusive projects: highest **NPV**, never the highest IRR.
- ! Profitability index = NPV / resource used; it ranks only under a constraint and can miss the best combination when projects cannot be split.
- ! Payback ignores discounting and everything after the cutoff.
- ! Growing perpetuity IRR = C_1/I_0 + g; a declining cash flow has a negative g.
- ! Unlevered net income uses the **marginal** tax rate and ignores interest.
- ! FCF subtracts the **change** in NWC, not its level; NWC is recovered at the end.
- ! Opportunity costs and cannibalization are included; sunk costs and unchanged overhead are not.
- ! Selling an asset: tax only the gain above book value.

Key terms

Capital budgeting — Deciding which investment projects (capital expenditures) a firm should accept.

NPV rule — Accept projects with NPV > 0; among alternatives take the highest NPV.

Internal rate of return (IRR) — The discount rate that makes NPV = 0; accept if IRR > cost of capital (only reliable for normal projects).

Delayed investment — Benefits come before costs; NPV rises with r and the IRR rule gives the wrong answer.

Multiple IRRs — Several discount rates give NPV = 0 when cash flows change sign more than once.

Payback period — Years needed to recover the initial investment; accept if below the cutoff.

Cutoff period — The pre-set maximum payback time.

ROCE / accounting rate of return	Average annual accounting profit ÷ investment; compared with a hurdle rate.
Mutually exclusive projects	At most one project can be chosen.
Profitability index	NPV ÷ resources consumed; a quick way to pick projects under a resource constraint.
Incremental earnings	Earnings with the project minus earnings without it.
Cannibalization	Sales of the new product displace sales of an existing product.
Opportunity cost	Value a resource would have provided in its best alternative use.
Sunk cost	A cost paid regardless of the decision; never included.
Unlevered net income	EBIT × (1 − τc), ignoring interest.
Free cash flow (FCF)	UNI + depreciation − CapEx − ΔNWC: incremental cash available to financiers.
Net working capital (NWC)	Current assets − current liabilities; an increase is a cash outflow.
Depreciation tax shield	Tax saved because depreciation is deductible: τc × depreciation.
Break-even analysis	Finding the input level at which NPV = 0.
Sensitivity analysis	Varying one assumption to see the effect on NPV.
Scenario analysis	Varying several assumptions at the same time.
Continuation (terminal) value	Value of cash flows after the explicit forecast period, e.g. a growing perpetuity at the end.